

Shaashvat Shetty

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Education

B.S. Computer Science | B.S. Statistics

University of Maryland, College Park

Expected Graduation: December 2026

GPA: 3.806 (CS Honors, Dean's List)

Relevant Coursework: Advanced Data Structures, Algorithms, Data Science & Machine Learning, Computer Systems, Statistical Computing with SAS, Applied Probability & Statistics

Experience

Research Assistant

August 2025 – Present

University of Maryland, College Park — Advisor: Prof. Haizhao Yang

- Researching the Finite Expression Method (FEX) to solve high-dimensional Partial Differential Equations using symbolic machine learning.
- Designing a model that projects high-dimensional equations into lower-dimensional latent space for faster and more efficient computations.
- Incorporating an LLM to assist in summarizing and dividing high-dimensional equations during the symbolic discovery process.
- Producing **explicit, interpretable finite expressions** as solutions, replacing black-box neural network approximations with transparent mathematical results.

Software Engineering Intern

June 2024 – September 2024

Juniper Networks, Sunnyvale, CA

- Built a Python-based network automation CLI that runs on a leaf switch in a CLOS architecture to validate EVPN-VXLAN connectivity and print per-connection status; reduced troubleshooting from hours to under 10 seconds.
- Implemented a second CLI command to explain failures by identifying likely misconfigurations (e.g., VLAN mismatch, interface issues).
- Validated the workflow across multi-leaf/spine topologies with hundreds of nodes, prioritizing reliability and clear terminal output for non-expert operators.
- Wrote maintainable code and documentation to support handoff and long-term ownership by the team.
- Developed unit/integration tests for CLI workflows and edge cases to improve reliability before deployment.

Computer Vision Intern

June 2022 – August 2022

UCSC SIP, Santa Cruz, CA

- Built an ML system for drone navigation from natural-language commands by combining object detection with text-to-object matching.
- Evaluated YOLOv5 vs. YOLOv3 and selected YOLOv3 for 35% faster inference on constrained hardware while maintaining similar accuracy across 60 classes.
- Improved robustness to synonym variations in commands (e.g., “bike” vs. “bicycle”) using cosine similarity without retraining.

Technical Skills

Languages: Python, SQL, C, Java, JavaScript, CSS, HTML, Bash, R, Rust, OCaml, SAS

Developer Tools: Git, GitHub, VS Code, Eclipse, Sublime, Jupyter Notebook, JUnit

Libraries: NumPy, Pandas, Matplotlib, scikit-learn, YOLOv3, PyTorch, TensorFlow, Keras

Technologies: Data Center Technology, EVPN-VXLAN, BGP protocols

Projects & Certifications

Custom iMessage Chatbot

July 2025

- Engineered a data pipeline to parse and structure 50k+ personal iMessages into a .JSONL format suitable for model training.
- Fine-tuned the TinyLlama-1.1B model (via Hugging Face) on the curated dataset to accurately learn and replicate a personal conversational style.
- Developed a lightweight interface (HTML, CSS, JS) that queries the model to predict and generate text replies mimicking my personal tone.

Analysis of Various Classification Methods on the Fashion MNIST Dataset

May 2025

- Benchmarked 7 classification models (e.g., SVM, CNNs) on the Fashion MNIST dataset to evaluate performance trade-offs.
- Implemented a full model pipeline, applying Principal Component Analysis (PCA) for dimensionality reduction and 5-fold cross-validation to ensure model robustness and prevent overfitting.
- Engineered a Convolutional Neural Network (CNN) that achieved 91.05% accuracy, confirming its superior performance over traditional models through a detailed comparative analysis using confusion matrices.

Map-Matching Algorithm Analysis

April 2025

- Investigated solutions to real-world GPS inaccuracies by implementing and rigorously evaluating four map-matching algorithms
- Developed synthetic GPS trajectory data incorporating controlled noise, simulating both dense urban and sparse rural conditions
- Found the Hidden Markov Model most accurate with mean error as low as 0.69 m